The Hidden Costs of Urban Renewal

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Check the paper: mateomoglia.github.io/files/JMPMoglia.pdf

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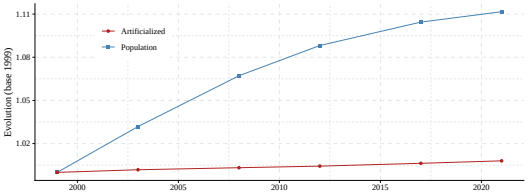
Large cities face a housing (supply) crisis

- Housing costs increase in large cities: +30% in the Paris area since the 2000s
- Same patterns elsewhere: Boston, NYC, San Francisco, London
- Housing supply crisis is fuelling political discontent and calls for action by local leaders



Redevelopment is the main source of new housing

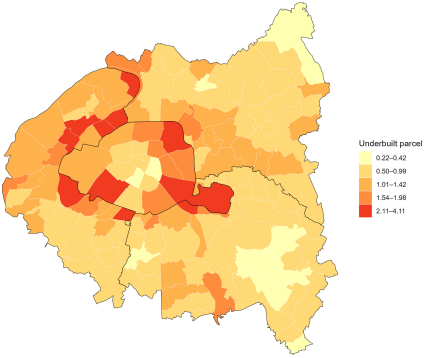
- Paris and its inner suburbs: **71%** of the units authorised since 2014 replace a building demolished on the same parcel
- 1999–2021: population +11%, built-up land +0.8%
  - Floorspace per resident: +3.9% since 2015



Housing supply in large cities is a matter of **redevelopment**

Redevelopment is low and slow

- **Low**: About 7,000 parcels are infills, empty, or vacant. If redeveloped: stock increase by 10%
- **Slow**: most idle parcels have not changed in a decade (eg, brownfields). From sale to completion, a project takes on average 4 years
- Supply responds little **and** late to high and rising rents (Saiz, 2010; Baum-Snow & Han, 2024; Rollet, 2025)



Why is redevelopment slow and low?

Three kinds of hidden costs stand between a parcel and a new building

	What	Would a planner face it?
Frictions	Irreversibility and uncertain rents, fixed cost, time to build	**Yes**: real costs of redeveloping
Market failure	Few developers buy land: markdown	**No**: a transfer that distorts the owner's decision
Policy wedges	Taxes and density caps	Policy choices

- Standard explanation: land-use regulations, but incomplete
- Make redevelopment **low** (fewer, smaller projects) or **slow** (projects start later)
- Costs do not call for the same policy: a friction calls for a transfer, a failure for a correction

Why market structure matters for measuring costs

- **Why measure the fixed cost?** It decides the policy: subsidy, market power correction, tax rebates, LUR
- Structural models recover **large fixed costs** from the slow pace of construction (Murphy, 2018; Peng, 2023; Rollet, 2025)
  - Every source of delay is attributed to the fixed cost
- Homebuilding is concentrated (Quintero, 2023; Favilukis et al., 2026; Lefebvre & Daly, 2026; Charreire et al., 2025)
  - With few buyers, owners of **built** parcels are paid less than the value of redeveloping them, so they sell later
- A model fitted to the pace of construction reads this delay as a **high fixed cost**

I measure the fixed cost from **land prices**, net of the markdown, taxes, and delays

This paper

Why is redevelopment slow, and what can policy do about it?

1. **Data**: a parcel-level panel of the Paris area, 2014–2025
   - 550,000 parcels: cadastre, universe of land sales and permits, 3D envelopes, LUR, taxes
2. **Model**: a real-options model of redevelopment with an **oligopsonistic land market**
   - Few developers post offers, owners hold the option and choose when to sell
3. **Measurement**: the price of land reveals the fixed cost, parcel by parcel, net of the other costs
   - The timing of redevelopment is not targeted, so it can be compared with the data. The model does not predict it
4. **Policy**: five policies one at a time, then the optimal mix
   - Instruments interact: the optimal mix is not the sum of the best policies

What I find

- **Frictions explain a large share of the measured shortfall**
  - Shapley shares of the 20-year shortfall: taxes and caps 52.2%, frictions 46.0% (fixed cost alone 19.5%), markdown 1.8%
  - Measured on the parcels whose price the model rationalises: 29% of parcels

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- **The market failure makes supply slow more than low, and blocks parcels elsewhere**
  - Owners receive 77% of what their land is worth to developers. Developers start projects on built parcels 3.5 years later than a planner would
  - Concentration blocks 1.5% of all unsold parcels in the data for ever, not only in the estimation sample
  - Ignoring the markdown would overstate fixed costs by 58%

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  - Ignoring the markdown would overstate fixed costs by 58%
- **Policies have different effects**
  - Lifting caps: +-2.01% stock at 20 years. Fixed-cost subsidy on high-cost parcels that pays for itself within 20 years: +4.48%. Tax rebates: +6.46%. Seller subsidy: +1.19%, 2.7 years saved. Inclusionary zoning with an offsetting subsidy: -1.16%
  - **The optimal mix**, within the bounds set by law, under balanced budget, lowers market rents by 6.82% on average over 20 years

Contribution

1. **Measuring the cost of building in built-up cities**: fixed costs from **land prices**, net of the markdown, not from the pace of construction

   Glaeser & Gyourko (2003), Gyourko & Molloy (2015), Murphy (2018), Peng (2023), Baum-Snow & Han (2024), Rollet (2025), Chapelle et al. (2023)

2. **Real options with a land market**: buyer power changes **when** owners sell, and separates **slow** from **low**

   Titman (1985), Capozza & Helsley (1990), Dixit & Pindyck (1994), Grenadier (1996), Lange & Teulings (2024), Lebret, Liu & Valentin (2026)

3. **Monopsony on the land market**: labour-market tools taken to land sales

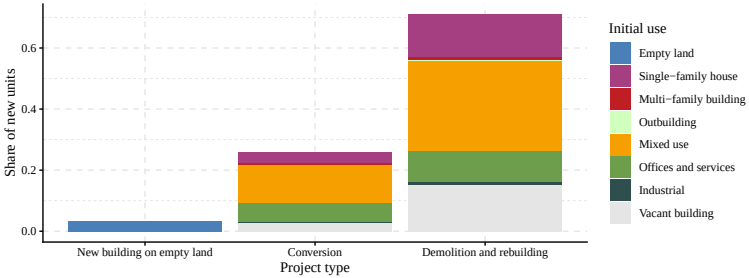
   Card et al. (2018), Berger, Herkenhoff & Mongey (2022), Quintero (2023), Lefebvre & Daly (2026)

4. **Ranking supply-side housing policies**: friction or failure, cost per unit, time saved

   Hsieh & Klenow (2009), Hsieh & Moretti (2019), Soltas (2024)

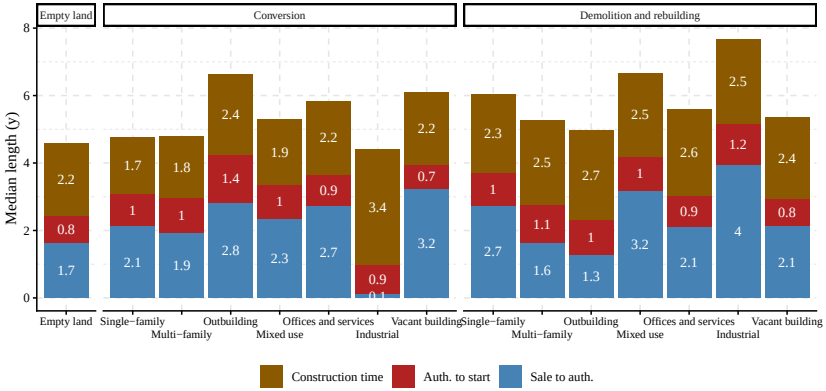
Data and facts

Housing is supplied by replacing what already stands



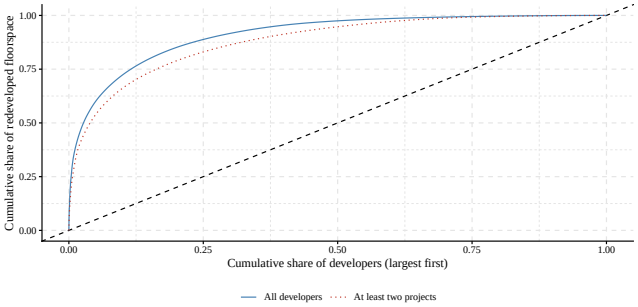
- 15,694 permits, 285,343 units, 16.5 million m^2 (2014–2024)
- **71%** of new units come from demolition and rebuilding, 3% from empty land
- Projects replacing offices are the densest: floor-area ratio 1.76, against 0.53 for houses

Friction: redevelopment takes years



- From the sale of the parcel to completion: 4.6 to 6.7 years Tests and lags
- Replacing offices, services, or industry takes longest

Failure: land is bought by a handful of developers



- **Fragmented across the region**: the top 7 groups supply 16% of redeveloped floorspace
- **Concentrated within municipalities**. In the median municipality:
  - 3.0 effective developer groups
  - The largest holds 50% of purchases

A lower bound: project companies are not all traced to their parent group

Failure: the municipality is the market

- **Developers buy locally**: 56% of a group's purchases are made in a single municipality
- **Not just small numbers**: a municipality has a median of 12 projects over three years
  - Benchmark: developer groups drawn at random across the projects of the region
  - 80% of municipalities are above the 95th percentile of their random HHI Table
- On the output side, no project moves rents: the largest is 1.35% of the municipal stock, and nearby prices do not react Donut

Price takers on housing, **price setters on land**

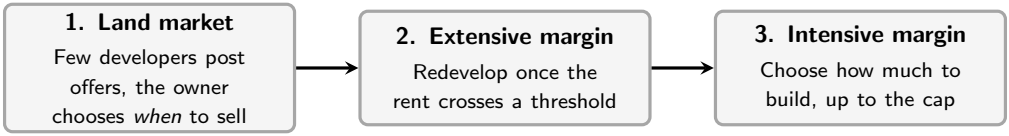
A model of redevelopment

What the model is for

- Uncover the **cost** of redeveloping each parcel: the fixed cost, net of the markdown, taxes and delays
- Predict the **timing**: when each parcel is redeveloped, privately and by a planner
- Compute the **optimal supply** parcel by parcel: what a planner would build, where, and when

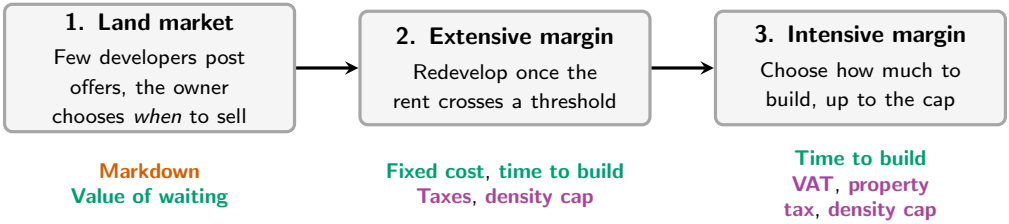
Each cost is then switched off, alone or with the others, and each policy is evaluated on the same parcels

The model in three steps



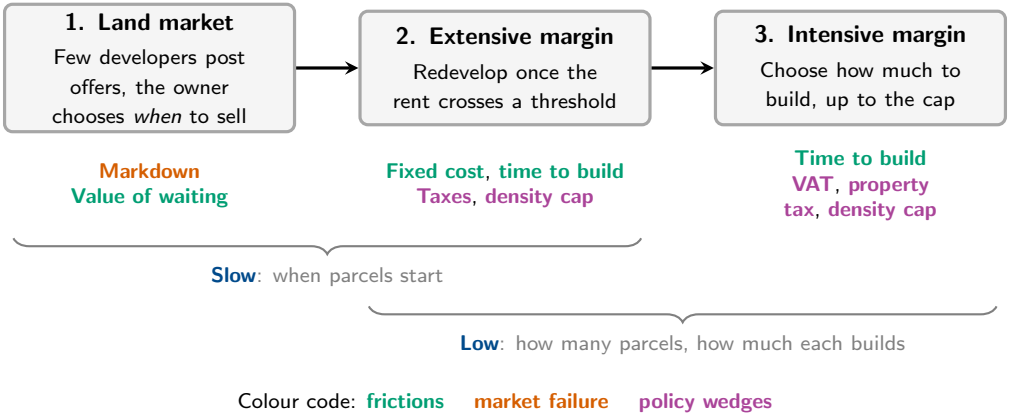
Colour code: **frictions** **market failure** **policy wedges**

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The model in three steps



Demand and rents in general equilibrium

- **Demand** A_t follows a geometric Brownian motion

$$\frac{dA_t}{A_t} = \mu\, dt + \sigma\, dW_t$$

- Rents clear the market, given the stock H_t

$$R_t = A_t H_t^{-\alpha}$$

- Parcels are small: agents expect rents to grow at δ, net of future supply

Why R_t follows an adapted GBM

- The stock moves slowly and without noise: it only grows when parcels are delivered
- So R_t inherits the shocks of A_t, and new supply lowers its drift

$$\frac{dR_t}{R_t} = \Big(\mu - \alpha\, \frac{d\ln H_t}{dt}\Big)dt + \sigma\, dW_t$$

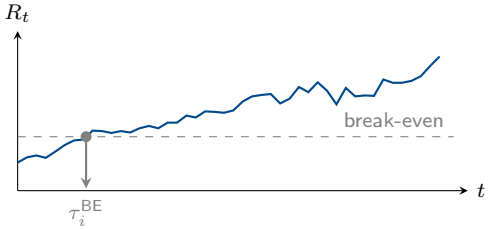
- Between deliveries, R_t is a GBM with the drift of demand

Each parcel is a price taker on housing. Supply feeds back on rents only through the stock H_t

Why wait? Uncertain rents, irreversible redevelopment

- A net-present-value rule would build at **break-even**, at τ_i^{BE}

  The rent at which the new building just pays for its cost and for what it replaces

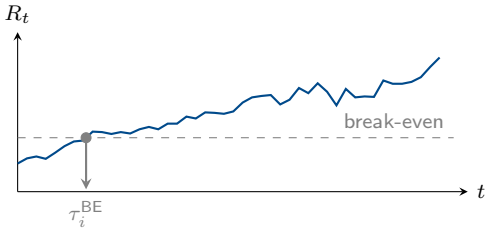


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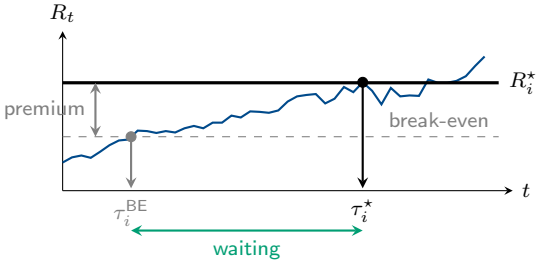
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- Redevelop only once the rent reaches a **threshold** $R_i^\star$ above break-even

$$R_i^\star = \underbrace{\frac{\beta}{\beta - 1}}_{\text{option premium}} \times R_i^{\text{break-even}}$$



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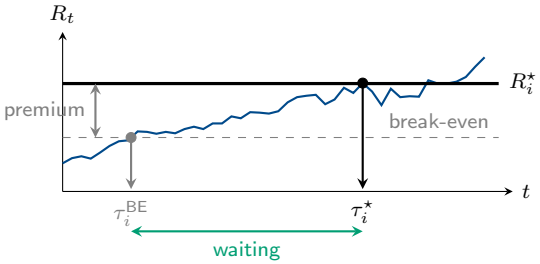
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- A planner waits too: the premium is a **friction**, here 1.48



Building today locks in today's rent. Waiting keeps the upside and drops the downside

Step 1. The land market: few buyers, owners paid a share of the value

- The developer value of parcel i: what the project is worth net of every cost

$$V_i^D(R) = \pi_i(R) - F_i - \text{taxes}$$

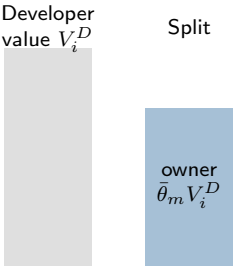
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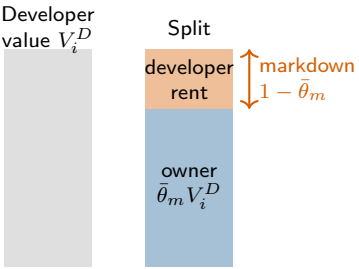
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- Buying one more parcel raises the offer on all: a **markdown** Proof

$$\frac{1}{\bar{\theta}_m} = 1 + \frac{HHI_m}{\eta^S} + \frac{1 - HHI_m}{\eta^D}$$



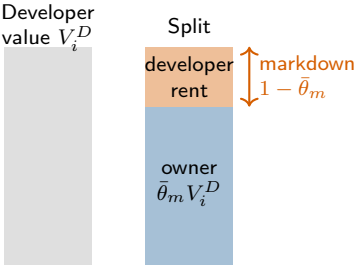
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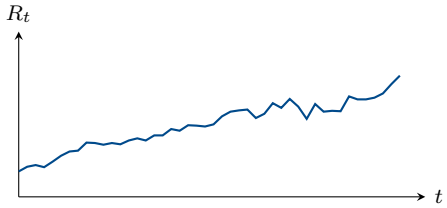


The land share $\bar{\theta}_m$ falls with local **concentration** HHI_m, measured by municipality

Why the municipality

The markdown makes owners wait: **slow**, and sometimes **low**

- The owner gives up the full standing value V_i^w but receives only $\bar{\theta}_m V_i^D$: **she sells once** $V_i^D > V_i^w / \bar{\theta}_m$

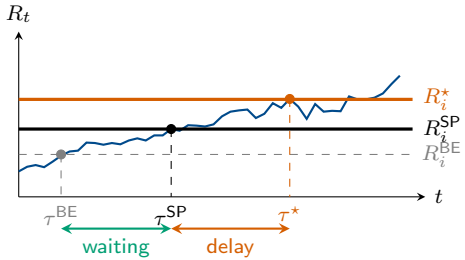


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$$R_i^{\text{BE}} \;<\; R_i^{\text{SP}} \;\leq\; R_i^\star$$

Break-even to planner: value of waiting.
Planner to private: markdown



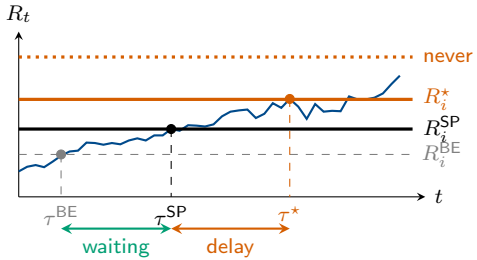
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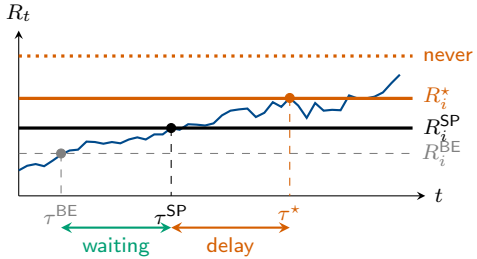
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Empty land: no distortion, $R_i^\star = R_i^{\text{SP}}$. A failure **of redevelopment**

Step 2. Extensive margin: redevelop once $R_t \geq R_i^\star$

- Most parcels build at the cap. The owner's surplus is then linear in the rent, and the owner sells at

$$R_i^\star = \underbrace{\frac{\beta}{\beta - 1}}_{\text{waiting}} \times (\rho - \delta) \times \frac{\overbrace{F_i + C_i + \text{land taxes} + V_i^w/\bar{\theta}_m}^{\text{what must be paid or given up}}}{\underbrace{\hat{\kappa}_i \bar{h}_i - h_i^{\text{old}}/\bar{\theta}_m}_{\text{net floorspace gained}}}$$

- V_i^w, h_i^{old}: value and floorspace of the standing building
- $\hat{\kappa}_i = e^{-(\rho-\delta)\ell_i}(1 - \hat{t}^F)/(1 + \tau^V)$: **aggregate cost shifter** of the **lag** and the **taxes** on the new building
- **Frictions**: $\beta/(\beta - 1)$, F_i, lag. **Failure**: $1/\bar{\theta}_m$. **Policy**: taxes, cap $\bar{h}_i$
- Every cost raises $R_i^\star$

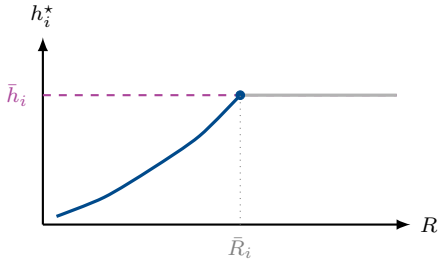
A higher **cap** raises both the construction cost C_i and the floorspace gained.
It can **delay** parcels where construction cost dominates: they wait to build taller

Step 3. Intensive margin: how much to build

- Once the parcel starts, the developer chooses density

$$h_i^\star(R) = \min\Big\{ \underbrace{T_i \Big[\frac{\hat{\kappa}_i \, R}{\text{construction cost}} \Big]^\zeta}_{\text{interior}}, \; \bar{h}_i \Big\}$$

- T_i: parcel area. ζ: elasticity of density to the rent (curvature of construction cost). $\bar{h}_i$: **zoning cap**

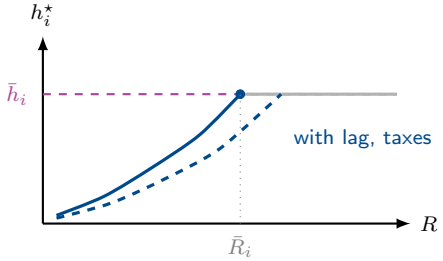


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- A lower cost shifter $\hat{\kappa}_i$ (**time to build**, **taxes**) lowers the interior density at a given rent

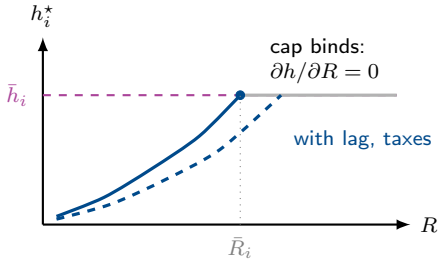


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- The **cap** binds for 92% of finite thresholds



Where each cost acts

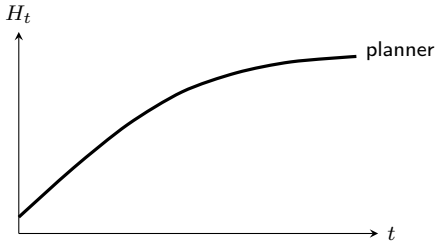
	Cost	Land market	Extensive margin	Intensive margin	Planner bears it
Frictions	Fixed cost F_i		✓		✓
	Value of waiting $\beta/(\beta-1)$	✓	✓		✓
	Time to build ℓ_i		✓	✓	✓
Failure	Markdown $1-\bar{\theta}_m$	✓	✓		
Policy	Taxes	✓	✓	✓	✓
	Density cap $\bar{h}_i$		✓	✓	✓

- **Only the planner's column separates the markdown from the fixed cost**
- Removing a friction is a transfer to be financed. Removing the markdown raises welfare on its own
- A subsidy to sellers can restore the planner's timing, using only the local HHI_m

From parcels to supply: slow versus low

- A parcel starts when the running maximum of rents crosses its threshold. It delivers ℓ_i years later

$$H_t = \bar{H} + \sum_u \Delta_u \big(\mathcal{M}_{t-\ell_u}\big), \quad R_t = A_t H_t^{-\alpha}$$

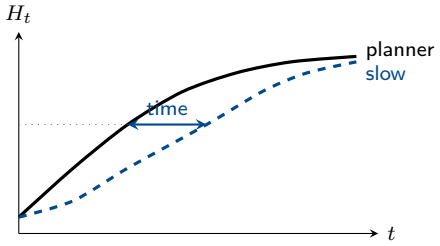


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- **Slow**: a cost shifts the path to the right

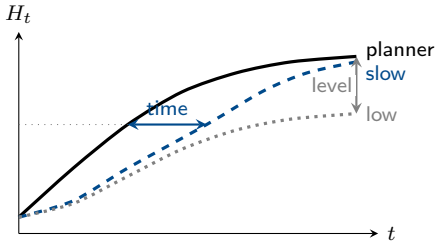


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- **Slow**: a cost shifts the path to the right
- **Low**: a cost lowers where the path ends



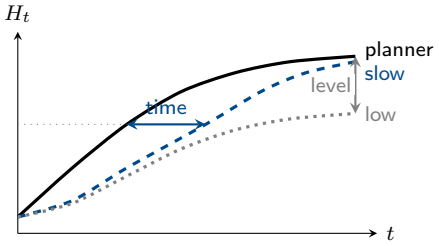
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- **Slow**: a cost shifts the path to the right
- **Low**: a cost lowers where the path ends

I report the change in the stock at 20 years, and the **years saved** to reach the stock of the private equilibrium at year 20



How large are the costs?

Calibration

1. **Set outside the model**
   - Yield $y = 3.5\%$, drift $\delta = 1.5\%$, demand volatility from transaction prices, $\sigma = 3.4\%$
   - Inverse demand elasticity, open city, $\alpha = 0.52$
   - Construction cost from a cost survey, curvature from Combes et al. (2021)
   - Lags: median durations by initial use. Taxes: statutory rates
2. **Markdown**
   - Slope: land sales followed by a permit, on the municipal HHI_m
   - Level: operating margins of listed developers
   - Owners receive 77% of the developer value

How large is the markdown?

- The slope is imprecise, $\varsigma \in$ [-2.29, 0.54]. I take the largest markdown the data allow
- The level sits at a corner: margins 8.0% against a target of 8.0%
- Project companies are not yet all traced to their parent group: measured concentration is **too low**

Parameters Validation

Recovering the fixed cost from land prices

- The price of an unsold parcel is its value to the owner [Land prices]

$$\text{Price}_i = \underbrace{\text{standing value}_i}_{\text{observed}} + \underbrace{\bar{\theta}_m}_{\text{land share}} \times \underbrace{\text{option to redevelop}_i(F_i)}_{\text{falls with } F_i}$$

- Given the land share, the price reveals the option, and the option reveals F_i, parcel by parcel
- The timing of redevelopment is not targeted. The model rationalises 29.2% of prices
  - Left out: prices below the standing value (13%), above the zero-cost bound (51%), blocked, top [Typology of parcels]
- **Ignoring the markdown** ($\bar{\theta}_m = 1$): the same prices imply fixed costs 58% higher
  - The delay due to the markdown is read as a cost

The fixed cost is a residual: clearing, co-ownership, tenants, heritage, assembly, etc. [What is in F_i]

How large is each cost?

Cost	Measure	Value
Fixed cost	Median F_i per m^2 of floorspace at the cap	592 € (0.50 $\times c_0$)
Value of waiting	Option premium $\beta/(\beta-1)$	1.48
Time to build	Lag from sale to delivery	4.6–6.7 years
Markdown	Share of developer value paid to owners	77% (P10–P90: 62–86%)
Taxes	VAT, transfer tax, development tax	20%, 6.32%, 88 €/m^2
Density cap	Share of thresholds where the cap binds	92%

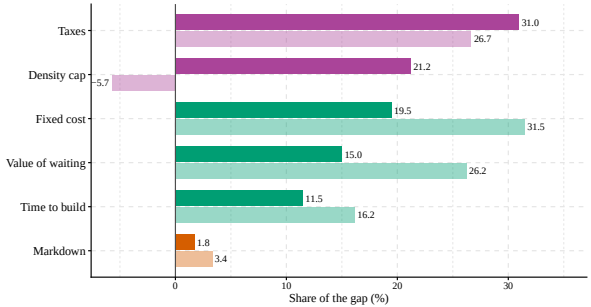
How much supply do the costs hold back?

Low: frictions explain most of the shortfall, the fixed cost first

- **The gap**, in quality-adjusted stock at 20 years

$$\text{Gap} = H_{20}^{\text{no wedge}} - H_{20}^{\text{private}}$$

- Remove every subset of the six wedges: $2^6 = 64$ bundles, same demand paths
- **Shapley share** of a wedge: its average contribution over every order of removal. Shares sum to the gap
- **Removed alone**: the wedge removed from the private equilibrium, all others kept

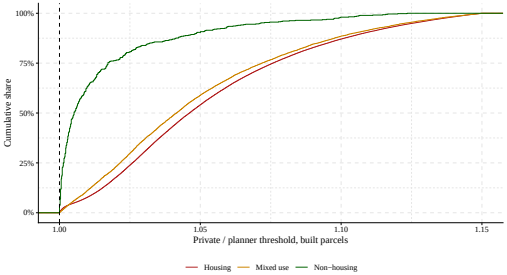


Table

Slow: the markdown delays more than it blocks

- The markdown has a small Shapley share of the level: 1.8%
- But it shifts timing: a built parcel starts **3.5 years later** than a planner would let it
- It also blocks some parcels for ever: 1.5% of all unsold parcels in the data
  - The new building is worth less than the standing one scaled up by $1/\bar{\theta}_m$

  Blocked parcels | By wedge | Typology
- It can also change the **order** in which parcels are redeveloped, though in the data it barely does Misallocation



Private over planner threshold, built parcels

Policies

Individual policies

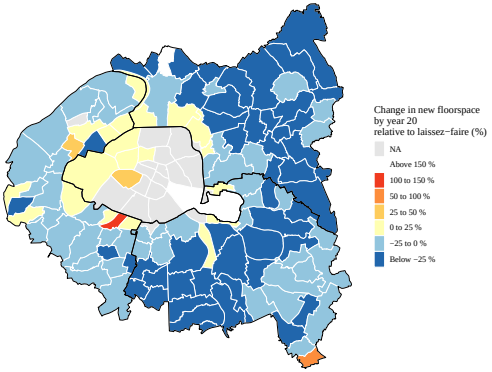
1. **Lift the density cap** to a floor-area ratio of 12.47 (1.5 $\times$ the 99th percentile of caps)
2. **Subsidise the fixed cost** of high-cost parcels (self-financing at 20 years)
3. **Rebate taxes** on redevelopment
4. **Subsidise sellers** to restore competitive timing
5. **Inclusionary zoning with an offsetting subsidy** Details

Reported for every policy, general equilibrium, same demand paths, 143,098 unsold parcels

- **Low**: change in stock and rent at 20 years
- **Slow**: years saved to reach the stock of the private equilibrium at year 20
- **Cost**: net public cost per additional unit, receipts counted over the same 20 years

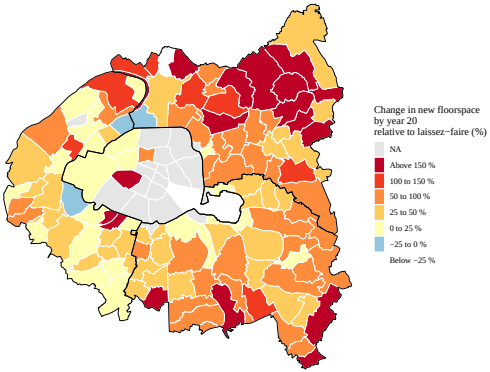
(1) Lifting the density cap

- Stock +-2.01%, rents —-1.04% after 20 years
- -3.0 years saved, -7.1 million m^2 more
- Costs nothing directly. Induced projects pay taxes: net cost -50.3 k€ per unit
- But it delays 87% of the parcels it reaches: they wait to build taller



(2) A fixed-cost subsidy that pays for itself

- A uniform subsidy never pays for itself: 0% of its outlay goes to parcels that start today anyway
- Pay the fixed cost above 250 €/m^2 of land, up to **800 €/m^2**
- Stock +4.48%, rents −2.24% after 20 years, 11.8 years saved
- Benchmark: the *Fonds Friches* scale, 50 €/m^2
- By year 50 it costs 61.5 k€ per unit: it brings projects forward more than it adds them



(3) Tax rebates

- Two instruments that already exist for some projects
  - VAT on new housing at the reduced rate of 5.5% (priority neighbourhoods, social housing)
  - No transfer tax on land bought to build, as already for buyers who commit to build within four years

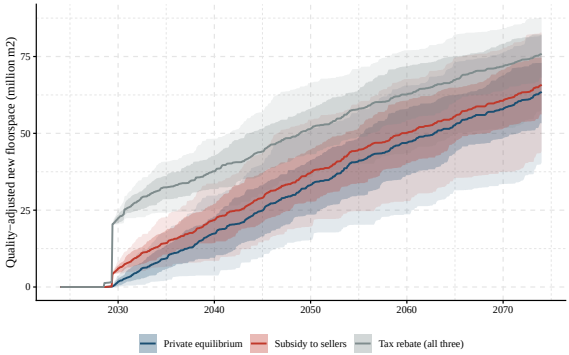
- No outlay: the cost is the receipts forgone, also on projects that would start anyway

	Stock (%)	Rent (%)	Years saved	Net cost/unit (k€)
VAT at 5.5%	+4.23	−2.13	10.8	15.4
No transfer tax	+0.37	−0.19	0.7	41.2
Both	+6.46	−3.18	13.8	69.5

Removing all taxes would close 26.7% of the gap to the economy without wedges. VAT goes to the State, the transfer tax to the départements

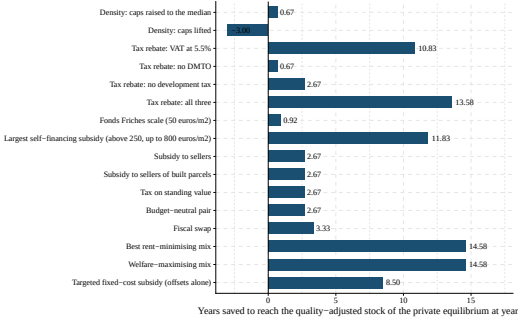
(4) Correcting the markdown: a subsidy to sellers

- Top up the price each seller receives: the seller gets the **full** value of the land
- Top-up per euro paid: $1/\bar{\theta}_m - 1$, larger where few developers buy. Owners then sell when a planner would
- Stock $+1.19\%$, **2.7 years saved**
- Costly: 75.6 k€ per unit, 0.9 $\times$ the cost of building one. 85% of the outlay goes to owners who would have sold anyway

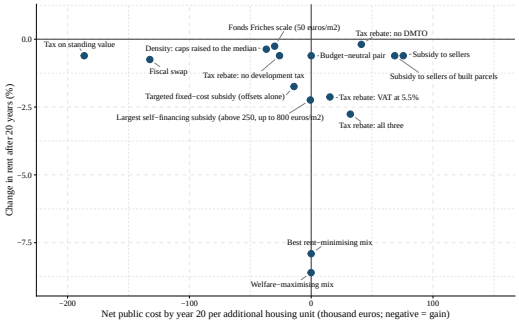


Taking stock of individual policies

Slow: years saved relative to the private equilibrium at year 20



Low and cost: rent change against net cost per unit

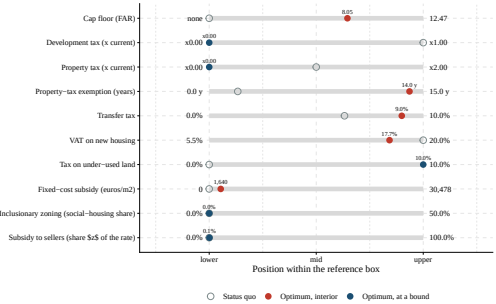


Caps are free and fast. Subsidies and rebates cost money. Only the seller subsidy corrects a failure

An optimal policy mix

An optimal policy mix

- 10 instruments. Minimise rent at 20y under balanced budget
- Every instrument within the bounds set by law or by the data
- Market rents −6.82% on average, stock +17.29%, 14.6 years saved
- 5 of 10 instruments end at a policy bound
- Instruments are complements: the mix beats the sum of its parts by 1.01 points
- Renters of the existing stock gain 59.3 bn €, paid by landlords



Each instrument within its policy bounds

Mix | Contributions

Within the bounds set by law: market rents −6.82% at a balanced budget

Conclusion

Conclusion

- Where cities grow inward, redevelopment is held back by **hidden costs**: frictions, a market failure, and policy wedges
- **Low**: frictions explain most of the measured 20-year shortfall, and the fixed cost, a residual, is the largest single cost (19.5%)
- **Slow**: concentration on the land market delays built parcels by 3.5 years, and blocks 1.5% of all unsold parcels in the data for ever. Ignoring it overstates fixed costs
- **Policy**: lifting caps is fast and cheap. A fixed-cost subsidy pays for itself within 20 years through VAT only if it skips low-cost parcels, which start anyway. Targeted parcel by parcel, it does 0.8 times more. A subsidy to sellers is the only policy that removes a failure, and it mostly makes supply faster
- Together, within the bounds set by law and under a balanced budget, they lower market rents by 6.82% on average over 20 years

The Hidden Costs of Urban Renewal

Matéo Moglia

CREST - IPParis

Check the paper: mateomoglia.github.io/files/JMPMoglia.pdf

Appendix

Construction lags by initial use

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Initial use	Sale to permit	Permit to start	Works	Sale to completion	Permits
Vacant	1.7	0.8	2.2	4.6	284
Housing	1.9	1.0	2.5	5.4	1,617
Mixed	3.2	1.0	2.5	6.7	1,896
Non-housing	2.7	1.0	2.5	6.2	1,030

- Lag ℓ_u (years): sum of the three median steps, from the sale of the parcel to completion

	Empty land	Outbuilding	Mixed	Multi	Industrial	Single	Offices
Outbuilding	0.170						
Mixed	0.050**	0.617					
Multi	0.617	0.156	0.034**				
Industrial	0.034**	0.700	0.248	0.034**			
Single	0.248	0.352	0.034**	0.110	0.077*		
Offices	0.007***	0.975	0.050**	0.004***	0.617	0.001***	
Vacant	0.110	0.617	0.617	0.050**	0.189	0.248	0.050**

Kruskal–Wallis: $\chi^2 = 32.7$, df $= 7$, $p = <0.001$, 2,910 permits

Calibration

Back

Par.	Name	Source	Value
μ, σ	Drift and volatility of demand (%)	Set by hand; transaction price index	1.5, 3.4
α	Inverse demand elasticity, open city	s_H, ε^L, ε^h	0.52
y, δ, ρ	Yield, perceived drift, discount rate (%)	Developers, rents, $y + \delta$	3.5, 1.5, 5.0
β	Root of the value of waiting	Computed	3.08
ζ, c_0	Cost curvature, construction cost	Combes et al. (2021), cost survey	1.17, 1,186
ℓ_u	Lag by initial use (years)	Permits, median durations	4.6–6.7
ϕ_m	Location premium, P10–P90	Rent index	0.90–1.54
η^D, η^S	Elasticities across developers, of selling	Land sales, margins	16.13, 1.48
HHI_m	Municipal HHI, median	Permit register	0.333
τ^M, τ^A	Transfer tax (%), development tax (€/m^2)	Tax code, planning code	6.32, 88

Validation

Back

Panel	Statistic	Value
A	Hit rate among parcels predicted to start in 2019-2024, bound at 2019 (%)	1.29
A	Base rate (%)	0.79
A	Parcels predicted to start (%)	7.01
B	Model supply elasticity	0.38
B	Supply elasticity, Paris area, Chapelle et al. (2023)	0.35
B	Model floorspace started, unsold stock, first 10 years (million m2/y)	1.64
B	... without the parcels ripe at the current rent (million m2/y)	1.64
B	Sitadel floorspace authorised, estimation sample (million m2/y)	0.20
B	Sitadel floorspace authorised, all parcels of the base data (million m2/y)	0.80
B	Sitadel floorspace authorised, all parcels of the area (million m2/y)	1.32
B	Model supply elasticity, scaled to observed flow	0.31
B	Unsold parcels ripe at the current rent (%)	0.00
B	Perceived drift delta = rho - y (%)	1.50
B	Realised drift mu - alpha g_Q, observed stock (%)	1.20
B	Realised drift of rents, simulated private equilibrium (%)	1.47
C	Model operating margin (%), aimed at the target	8.00
C	Listed developers (%)	8.00
D	Reduced-form slope of log land share on HHI	-0.52
D	s.e.	0.51
D	Model-implied slope	-0.52
E	Elasticity of sales to offers, parcels ripe at the current rent	–
E	Elasticity of sales to offers, starts by year 20	0.35
E	Elasticity of selling eta_S (calibrated)	1.48
F, HHI quintile 1	Priced above the zero-cost bound (%)	40.35
F, HHI quintile 2	Priced above the zero-cost bound (%)	66.42
F, HHI quintile 3	Priced above the zero-cost bound (%)	71.45
F, HHI quintile 4	Priced above the zero-cost bound (%)	64.54
F, HHI quintile 5	Priced above the zero-cost bound (%)	55.34

Untargeted checks

- Timing: parcels ranked ripe are bought at 1.3%, against a base rate of 0.8%
- Supply elasticity: 0.38 in the model, 0.5 in Chapelle et al. (2023)
- Realised drift of rents: 1.5% against $\delta = 1.5\%$
- Developer margin 8.0% against a target of 8.0%, missed at the corner

The land share: slope on concentration

Back

Statistic	Value
Slope of log land share on HHI, b	-0.524
s.e.	0.508
First stage (Wald)	40.0
Mean HHI	0.173
Implied ς (calibration)	-0.577
s.e.	0.614
Implied ς, 95% lower bound	-2.293
Implied ς, 95% upper bound	0.538
Land sales	1,590
Clusters (EPT)	12
Wild cluster bootstrap p-value, Anderson-Rubin test of $b = 0$	0.471
Wild cluster bootstrap p-value, 2SLS	–
b with EPT and year effects	-0.735
s.e.	0.545
b by OLS	-0.542
s.e.	0.218

- Sales of housing parcels followed by a permit. Instrument: average HHI of the other municipalities of the same territory
- Calibrated at the lower bound of the confidence interval: the largest markdown the data allow

What is in the fixed cost?

Back

Shapley decomposition of $\log F_i/T_i$

Block	All blocks (%)	Within municipality (%)
Demolition	9.46	14.39
Ground	16.27	9.27
Networks	0.00	0.02
Neighbours	12.93	19.26
Ownership	7.83	7.47
Area	13.60	37.49
Regulation	11.14	12.09
Municipal	28.77	–
R^2	0.10	0.07

Levels

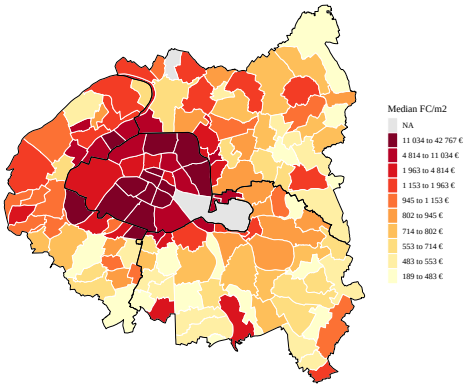
Block	Median share of F_i (%)	Aggregate share (%)	Median euros per m
Demolition	5.75	-13.85	27.24
Ground	-13.21	-6.28	-54.72
Networks	0.00	-0.01	0.00
Neighbours	0.73	14.28	2.52
Ownership	0.00	9.84	0.00
Area	-0.00	24.69	-0.00
Regulation	1.51	6.42	7.43
Municipal	0.19	-8.15	0.63

- Blocks: demolition (predicted from the cost survey), ground, networks, neighbours, ownership, area, regulation, municipality
- Municipal fixed effects explain 16% of the variance of $\log F_i/T_i$. Observables explain little: R^2 of 0.10 overall, 0.07 within municipalities

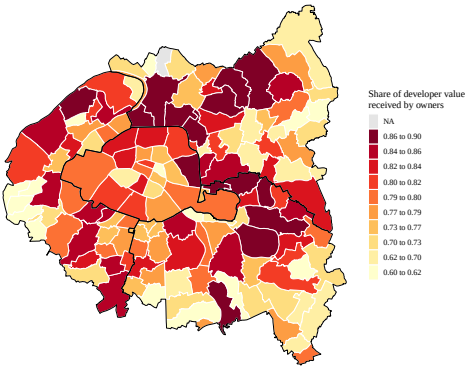
Fixed costs and land shares in space

Back

Fixed cost



Land share $\bar{\theta}_m$



The supply shortfall: Shapley table

Back

Wedge	Group	Shapley share of the gap (%)	Removed alone (% of the gap)
Markdown ($\bar{\theta}_m \to 1$)	Market failure	1.8	3.4
Option value	Friction	15.0	26.2
Time to build	Friction	11.5	16.2
Fixed cost	Friction	19.5	31.5
Taxes	Policy	31.0	26.7
Density cap	Policy	21.2	-5.7
Market failure	Total	1.8	3.4
Frictions	Total	46.0	44.7
Policy wedges	Total	52.2	43.0

Parcels blocked for ever by the markdown

Back

- The owner gives up the standing building in full but receives only the share $\bar{\theta}_m$ of the new one
- If the new building is not worth more than the standing one **scaled up by** $1/\bar{\theta}_m$, the owner never sells

$$\text{blocked:} \quad h_i^{\text{old}} \;<\; \hat{\kappa}_i \bar{h}_i \;\leq\; h_i^{\text{old}}/\bar{\theta}_m$$

- A planner would redevelop these parcels: the new building has more floorspace than the old one
- 7,158 parcels, 1.5% of all unsold parcels in the data, are blocked by concentration
  - Their private option is worthless, so their price reveals no fixed cost
  - They are outside the Shapley shares and outside the stock of the counterfactuals: the effect of the markdown is understated
- With depreciation, they would be delayed for very long rather than for ever

Which unsold parcels are held back by what

Back

Row	Parcels	Share of the unsold stock (%)	Share of potential floorspace (%)	Median delay due to the markdown (y)
Not ripe without wedges	336	0.2	0.2	9.4
Held back by markdown only	0	0.0	0.0	–
Held back by frictions only	94,930	66.3	68.1	3.3
Held back by taxes+caps only	0	0.0	0.0	–
Held back by several wedges	47,832	33.4	31.7	4.1
Ripe at the current rent	0	0.0	0.0	–
Memo: blocked by concentration, all unsold parcels	7,158	1.5	–	–

- **Frictions** alone hold back the parcels with most of the potential floorspace. The **markdown** alone holds back none
- A third of unsold parcels face several wedges at once: no single policy releases them
- Parcels blocked by concentration come on top of these rows: they never sell under the markdown

Misallocation: which parcels go first

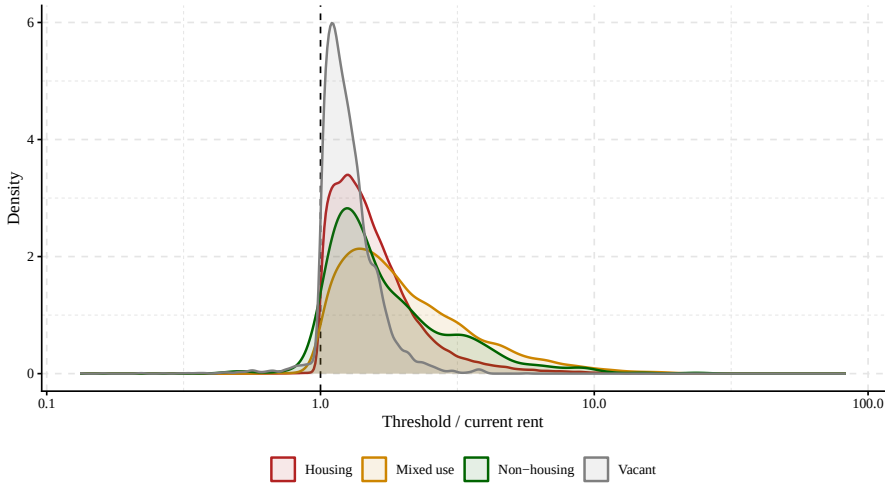
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Wedge	Med. wedge	IQR	Corr. with ϕ_m	Rank corr. of thresholds	Floorspace built out of order (%)	Allocation share of the gap (%)	ϕ of new floorspace, with wedge	ϕ of new floorspace, without
Markdown	0.05	[0.03, 0.09]	-0.12	0.98	20.97	-1.57	1.01	1.00
Taxes	0.46	[0.40, 0.53]	-0.17	0.97	23.62	-0.42	1.01	1.00
Density cap	-0.17	[-0.20, -0.10]	0.10	0.95	47.82	–	1.01	1.02

- A uniform wedge only delays. A dispersed wedge reorders parcels
- Markdown wedge: median 0.05, correlation with the location premium -0.12
- Resource cost at equal output, costed at the same date: -2.4%
- Rank correlation of private and planner thresholds: 0.98. In practice the markdown delays more than it reorders

Thresholds relative to the current rent

Back



All policies, 20 and 50 years

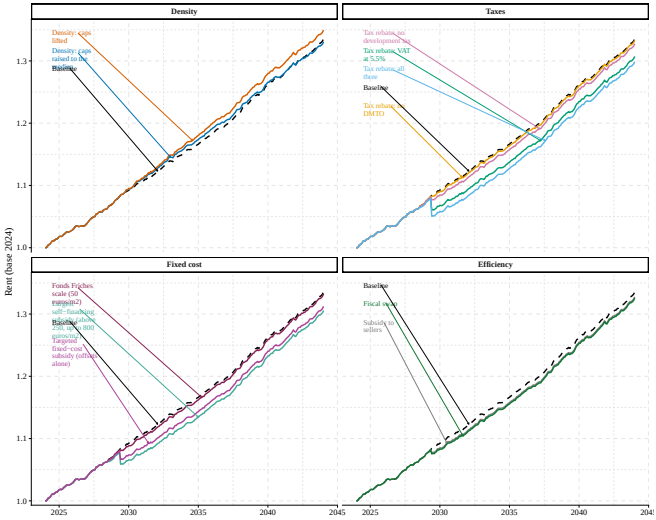
Back

Scenario	dH (%)	dR (%)	Net cost/unit (k€)	Time saved	Loc. premium	MVPF, narrow	MVPF, broad
Density: caps raised to the median	0.61	-0.37	-36.88	0.67	1.02	∞	∞
Density: caps lifted	-2.01	1.04	–	-3.00	1.02	∞	∞
Tax rebate: VAT at 5.5%	4.23	-2.13	15.38	10.83	1.00	1.51	1.56
Tax rebate: no DMTO	0.37	-0.19	41.16	0.67	1.01	1.28	1.28
Tax rebate: no development tax	1.18	-0.61	-26.01	2.67	1.01	15.12	15.35
Tax rebate: all three	5.56	-2.76	32.12	13.58	1.00	1.42	1.47
Fonds Friches scale (50 euros/m2)	0.50	-0.26	-29.99	0.92	1.01	15.17	15.28
Largest self-financing subsidy (above 250, up to 800 euros/m2)	4.48	-2.24	-0.70	11.83	1.00	2.47	2.59
Subsidy to sellers	1.19	-0.61	75.60	2.67	1.00	1.15	1.16
Subsidy to sellers of built parcels	1.19	-0.61	68.58	2.67	1.00	1.19	1.19
Tax on standing value	1.19	-0.61	-186.37	2.67	1.00	–	–
Budget-neutral pair	1.19	-0.61	-0.00	2.67	1.00	2.54	2.56
Fiscal swap	1.46	-0.75	-132.49	3.33	1.00	–	–
Best rent-minimising mix	17.29	-7.91	-0.02	14.58	0.99	2.77	3.19
Welfare-maximising mix	19.01	-8.60	-0.10	14.58	0.98	3.23	3.77
Targeted fixed-cost subsidy (offsets alone)	3.44	-1.74	-14.13	8.50	1.00	4.98	5.18

Scenario	dH (%)	dH (50y, %)	dR (%)	dR (50y, %)	Net cost/unit (k€)	Net cost/unit, (50y, k€)
Density: caps raised to the median	0.61	5.88	-0.37	-3.02	-36.88	-42.27
Density: caps lifted	-2.01	7.37	1.04	-3.97	–	-27.92
Tax rebate: VAT at 5.5%	4.23	2.62	-2.13	-1.26	15.38	97.16
Tax rebate: no DMTO	0.37	0.19	-0.19	-0.09	41.16	201.44
Tax rebate: no development tax	1.18	0.52	-0.61	-0.24	-26.01	7.80
Tax rebate: all three	5.56	3.12	-2.76	-1.48	32.12	125.47
Fonds Friches scale (50 euros/m2)	0.50	0.14	-0.26	-0.06	-29.99	10.91
Largest self-financing subsidy (above 250, up to 800 euros/m2)	4.48	1.62	-2.24	-0.71	-0.70	61.53
Subsidy to sellers	1.19	0.60	-0.61	-0.29	75.60	285.26
Subsidy to sellers of built parcels	1.19	0.60	-0.61	-0.29	68.58	265.08
Tax on standing value	1.19	0.60	-0.61	-0.29	-186.37	-439.23
Budget-neutral pair	1.19	0.60	-0.61	-0.29	-0.00	76.19
Fiscal swap	1.46	0.72	-0.75	-0.35	-132.49	-294.17
Targeted fixed-cost subsidy (offsets alone)	3.44	1.93	-1.74	-0.90	-14.13	-29.90

Mean rent paths by policy

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The seller subsidy: all sellers or built parcels only

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Scenario	dH (%)	dR (%)	Net cost (m€)	Net cost/unit (k€)	Time saved	Add. floorspace (m m2)	Loc. premium	Inframarginal (%)	Change in owners' value (m€)
Subsidy to all sellers	1.19	-0.61	4,657.50	75.60	2.67	4.25	1.00	85.20	11,904.99
Subsidy to sellers of built parcels	1.19	-0.61	4,225.53	68.58	2.67	4.25	1.00	84.49	11,324.20
Tax on standing value	1.19	-0.61	-11,482.24	-186.37	2.67	4.25	1.00	–	-5,509.81
Budget-neutral pair (z = 71%)	1.19	-0.61	-0.00	-0.00	2.67	4.25	1.00	85.20	6,879.54

- Rate $1/\bar{\theta}_m - 1$, 32% on average. Needs only the local HHI
- Costly: 85% of the outlay goes to owners who would have sold anyway

Inclusionary zoning with an offsetting subsidy

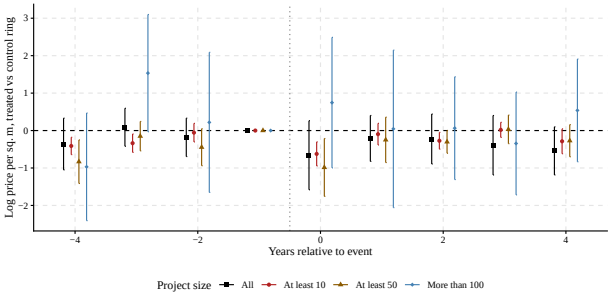
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Policy	dH (%)	dR (%)	Social floorspace (thousand m2)	Private m2 forgone/social m2	Outlay (m€)	Receipts / outlay (%)
Inclusionary zoning only	-4.05	2.21	2,725.32	7.17	0.00	–
Offsetting subsidy only	3.44	-1.74	0.00	–	10,301.87	124.08
Inclusionary zoning with an offsetting subsidy	-1.16	0.61	5,655.64	1.90	5,703.57	-107.52

- 45% of municipalities are short of their social-housing target
- 25% of new floorspace at half the market rent: inclusionary zoning alone lowers the stock by 4.05%

Prices around large projects

Back



- Donut difference-in-differences around projects: sales at 20–100 m against 150–500 m
- Event: year of the permit
- Effect: −0.056 (s.e. 0.169), minimum detectable effect 47.4%
- 40 projects, 2,648 sales

The land share: derivation

Back

1. Owners sell L_{jm} to developer j at offer θ_{jm}. Private preferences over selling (η^S) and over buyers ($\eta^D > \eta^S$) give nested CES supply

$$L_{jm} = \left(\frac{\theta_{jm}}{\Theta_m}\right)^{\eta^D} \left(\frac{\Theta_m}{\bar{\Theta}}\right)^{\eta^S} \bar{L}_m$$

2. Developer j maximises $(1-\theta_{jm})L_{jm}$ with one offer to all owners. The inverse supply elasticity it faces is

$$\frac{\partial \ln \theta_{jm}}{\partial \ln L_{jm}} = \frac{w_{jm}}{\eta^S} + \frac{1-w_{jm}}{\eta^D}, \qquad w_{jm} \text{ : payment share}$$

3. First-order condition: $1/\theta_{jm} = 1 + w_{jm}/\eta^S + (1-w_{jm})/\eta^D$

4. Aggregate with payment weights: $1/\bar{\theta}_m = \sum_j w_{jm}/\theta_{jm} = 1 + HHI_m/\eta^S + (1-HHI_m)/\eta^D$. Exact

Why the municipality is the land market

Back

- **Developers buy locally**: 56% of a group's purchases are made in a single municipality
- **Permits and zoning are municipal**: the mayor signs the permit and sets the local plan. Developers build local networks and negotiate programmes municipality by municipality
- **Owners face local buyers**: the developers that prospect a parcel are those active in its municipality
- **Not just small numbers**: concentration exceeds random allocation of groups across projects Table
- The municipality also closes the model: one land share per local market, one HHI_m per market

Local concentration

Back

Municipalities	N	Median HHI, observed	Median HHI, random allocation	Above the 95th percentile of random allocation (%)
All with a permit in the last window	137	0.33	0.20	80
Fewer than 10 projects	55	0.59	0.37	64
10 to 19 projects	39	0.31	0.17	90
20 projects or more	43	0.20	0.10	91

- Random allocation: developer groups permuted across the projects of the region, 500 draws

Who is in the estimation sample

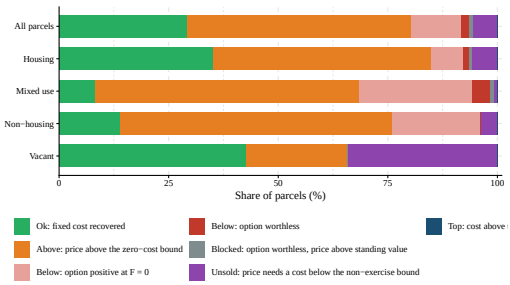
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Statistic	All parcels	Unsold	Sold by 2025	Out: price below standing value	Out: price above the zero-cost bound	Out: unsold, cost below the non-exercise bound	Out: option worthless or cost at the bracket
Parcels	498,257	143,098	2,162	65,787	255,725	27,805	3,680
Housing (%)	73	88	59	47	70	74	71
Mixed (%)	23	6	15	52	27	3	28
Non-housing (%)	1	0	2	1	1	1	0
Vacant (%)	4	5	25	0	2	22	0
Vacant (number)	18,315	7,312	538	0	4,221	6,244	0
Area (m2), median	323	399	452	333	271	383	188
Standing FAR, median	0.39	0.24	0.04	1.24	0.48	0.13	1.23
Cap FAR, median	1.74	1.39	1.81	3.16	2.28	1.06	1.35
Location premium, median	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Price per m2 of land, median	1,873	977	1,029	2,265	2,994	1,178	5,032
Paris (%)	12	0	1	16	19	0	21

- 145,260 parcels with a recovered fixed cost, of which 143,098 unsold: the stock of every counterfactual

A typology of parcels: what land prices say

Back



- **Ok** (29%): price between the standing value and the option at zero cost. A unique F_i fits. Mostly vacant or lightly built
- **Below** (13%): price at or below the standing value. Option worthless for 15% of them, under 5% of the standing value for 70%. Their fixed cost barely matters for supply
- **Above** (51%): price above the option at zero cost. Only a negative cost fits. Upper bound in the robustness table
- **Blocked**: price above the standing value, option worthless at zero cost
- **Top**: the fixed cost would exceed the bracket of the inversion

Only ok parcels enter the counterfactuals. Blocked and top are small. Parcels blocked by concentration Definition

Land prices: operations, not deeds

Back

- A site is often bought in several deeds: two flats in March, the other flats and the land in April
  - The April deed alone understates the price of the site
- I sum the purchases of the same buyer on the same site in the five years before a housing permit
  - 3601 operations, 16.5 % with several deeds
  - With several deeds, the operation costs 2.73 times its last deed
- Hedonic PPML on complete operations, municipality and year effects
  - Housing model for housing parcels, pooled built model for mixed and non-housing, vacant model for vacant
  - Out-of-sample predicted/observed: housing 0.97 , mixed 0.91 , non-housing 0.84 , vacant 1.23
- Assembly premium: elasticity 0.01 to the number of parcels assembled
  - Parcels are priced as single-parcel sites: the holdout cost stays in F_i

A tax on under-used land

Back

Tax rate (%)	dH, 20 y (%)	dR, 20 y (%)	Time saved (y)	Unsold parcels redeveloped by year 20	Revenue, years 1-10 (m€/yr)	Revenue, years 11-20 (m€/yr)
1.00	0.22	-0.11	0.33	1,153.43	75.79	72.04
2.00	0.43	-0.22	0.83	2,260.13	148.80	141.24
5.00	1.04	-0.53	2.25	5,414.90	353.20	332.95

The optimal mix

Back

Lever	Status quo	Policy bounds	Optimum	At a bound
Cap floor (FAR)	none	none – 12.47	8.05	no
Development tax (x current)	x1.00	x0.00 – x1.00	x0.00	yes
Property tax (x current)	x1.00	x0.00 – x2.00	x0.00	yes
Property-tax exemption (years)	2.0 y	0.0 y – 15.0 y	14.0 y	no
Transfer tax	6.3%	0.0% – 10.0%	9.0%	no
VAT on new housing	20.0%	5.5% – 20.0%	17.7%	no
Tax on under-used land	0.0%	0.0% – 10.0%	10.0%	yes
Fixed-cost subsidy (euros/m2)	0	0 – 30,478	1,640	no
Inclusionary zoning (social-housing share)	0.0%	0.0% – 50.0%	0.0%	yes
Subsidy to sellers (share z of the rate)	0.0%	0.0% – 100.0%	0.1%	yes
Average rent paid, change, mean over 20 years (%)	0.00		-6.82	
Market rent, change, mean over 20 years (%)	0.00		-6.82	
Market rent, change at 20 years (%)	0.00		-7.91	
Stock change at 20 years (%)	0.00		17.29	
Net public cost (m€)	0		-17	
Time saved (years)	0.0		14.6	
Social floorspace (thousand m2)	0		0	

Policy bounds: set by law or by the support of the data. At a bound: within 1% of either end of its range

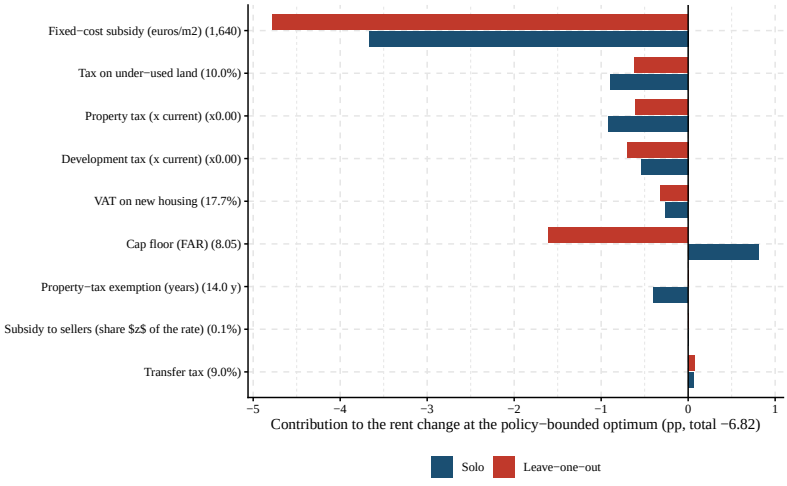
The optimal mix: fiscal accounts

Back

Account	Item	Billion euros
Public budget	Development tax	-2.53
Public budget	DMTO	7.19
Public budget	VAT	42.18
Public budget	Property tax, redeveloped parcels	-1.08
Public budget	Property tax, standing structures	-4.39
Public budget	Tax on under-used land	4.59
Public budget	Tax on standing value	0.00
Public budget	Change in fiscal revenue	45.96
Public budget	Fixed-cost subsidy	45.92
Public budget	Subsidy to sellers	0.02
Public budget	Outlays	45.94
Public budget	Net balance	0.02
Unsold parcels	Landowners	47.63
Unsold parcels	Developers	17.73
Housing market	Renters, lower rents on the stock	59.34
Housing market	Renters, surplus on additional housing	5.94
Housing market	Landlords of the existing stock	-59.34

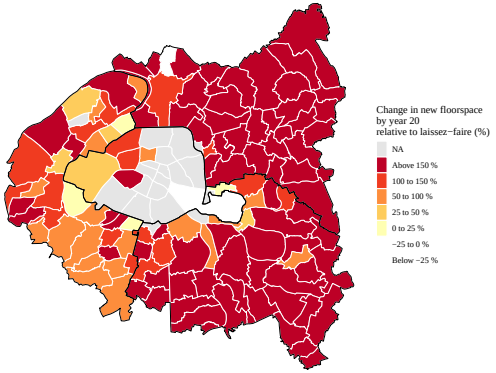
The optimal mix: contribution of each instrument

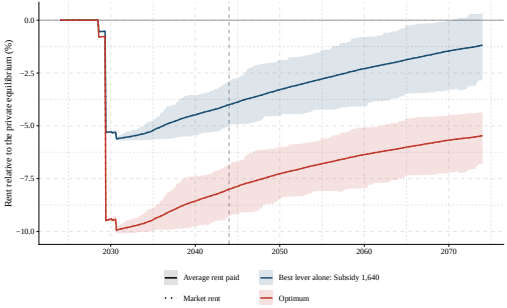
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The optimal mix: where it builds

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Robustness

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Variant	ρ (%)	y (%)	y_w (%)	δ (%)	σ (%)	β	Premium $\beta/(\beta-1)$	θ^{atom} (%)	Mean $\bar{\theta}_m$ (%)	Unsold parcels with a fixed cost	Blocked by concentration	Median F_i/T_i (€/m²)	Paris / suburbs, median F_i/T_i
Baseline	5.00	3.50	5.50	1.50	3.44	3.08	1.48	94.16	76.89	143,098	7,158	826.25	17.49
y + 50 bp, δ fixed	5.50	4.00	5.50	1.50	3.44	3.36	1.42	94.16	76.89	63,429	15,118	421.29	27.32
y - 50 bp, δ fixed	4.50	3.00	5.50	1.50	3.44	2.80	1.56	94.16	76.89	234,438	3,309	1,666.20	6.16
δ + 50 bp, y fixed	5.50	3.50	5.50	2.00	3.44	2.62	1.62	94.16	76.89	168,958	7,158	1,055.51	12.86
δ - 50 bp, y fixed	4.50	3.50	5.50	1.00	3.44	3.85	1.35	94.16	76.89	108,270	7,158	686.99	15.01
y_w + 50 bp	5.00	3.50	6.00	1.50	3.44	3.08	1.48	94.16	76.89	139,590	4,857	776.80	16.64
y_w - 50 bp	5.00	3.50	5.00	1.50	3.44	3.08	1.48	94.16	76.89	143,721	11,160	892.32	10.89
Slope ς at its 95% lower bound (-2.29)	5.00	3.50	5.50	1.50	3.44	3.08	1.48	99.85	54.91	72,708	44,810	715.01	20.24
Slope $\varsigma = 0$	5.00	3.50	5.50	1.50	3.44	3.08	1.48	82.78	82.78	157,218	4,546	844.91	16.17
$\sigma = 0.6\%$, rent index	5.00	3.50	5.50	1.50	0.64	3.32	1.43	94.16	76.89	132,214	7,158	771.26	16.56
Parcels priced above the zero-cost bound at $F = 0$	5.00	3.50	5.50	1.50	3.44	3.08	1.48	94.16	76.89	395,889	7,158	826.25	17.49